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SC-LFENM with energy bookkeeping

Scratch work, 2026-08-21. Nothing here is tracked, and nothing touches the `penm` package — it is a standalone testbed built to answer one question.

The question

You proposed this:

ENMs fit a network to a structure — given the native structure, they build completely relaxed springs, $l_{ij} = d_{ij}^e$. So it is consistent to *rebuild* the mutant’s network once its structure is known. But if I do that, I think I need to save the effect of the pre-built stress terms in order to calculate the mutant’s energy.

I set out to check whether the saving is really needed, what exactly has to be saved, and whether the resulting model works for the two things you want to do with it: single-mutation scanning and evolutionary trajectories.

Call the model SC-LFENM (self-consistent LFENM): mutate the natural lengths as in LFENM, relax, then *rebuild* a fully relaxed network on the relaxed structure, and repeat.

What I found.

1. The saving is needed, and what must be saved is one scalar per state (§§2–3). The rebuild erases the strain, so a mutation that genuinely costs 0.6 energy units is otherwise reported as costing zero.
2. For scanning, SC-LFENM works well (§7) and gives what LFENM cannot: a changed spectrum, hence mutational effects on dynamics.
3. The obvious way of running a trajectory fails (§6), and for a structural reason: re-fitting the network at every step makes the current state the zero of energy, so every move is uphill and the walk is absorbing. No amount of tuning selection repairs it.
4. Your catalogue fixes it (§8). Reference every mutation to the *founder* rather than to the current state. The energy then has a fixed floor and the state space is finite, so trajectories reach a genuine stationary state instead of stalling. The bookkeeping is exactly reversible and ΔV takes both signs. The price is a surrogate energy, additive over sites: accurate to $\sim 2\%$ in aggregate, but with per-move errors that can flip a sign, and degrading with distance from the founder, so the catalogue needs periodic rebuilding.
5. Frustration is created by every mutation and changes the Hessian by 0.6–2.7 %, but rotates the soft modes by less than 0.2 % (§9).
6. K_{wt} vs K_{mut} for the response — open since 2020 — is settled: the direction agrees to $\cos > 0.994$, only the magnitude differs (§10).

Notation. $\beta = 1/kT$ is Boltzmann’s, used only for thermodynamic quantities. Selection strength is written ν , a separate letter, because it is an evolutionary parameter and not a temperature.

1. Setting up: the potential, and three Hamiltonians

The potential

An elastic network model assigns to a conformation $\mathbf{r}$ the energy

$$V(\mathbf{r}; \{l_{ij}\}) = \sum_{i < j} \left[V_{ij}^{\min} + \frac{1}{2} k_{ij} (d_{ij}(\mathbf{r}) - l_{ij})^2 \right] \quad (1)$$

where $d_{ij}(\mathbf{r}) = \|\mathbf{r}_j - \mathbf{r}_i\|$ is the distance in the conformation $\mathbf{r}$, k_{ij} the force constant, and l_{ij} the natural (rest) length of the spring $i \leftrightarrow j$. Sums run over the contact set. Throughout I take $V_{ij}^{\min} = 0$ and keep the constant separately (§3).

Two arguments, and they play different roles. $\mathbf{r}$ is a variable — it ranges over conformations. $\{l_{ij}\}$ are parameters — fixing them fixes the Hamiltonian. Changing $\{l_{ij}\}$ does not move the protein; it replaces the potential by a different one.

The three Hamiltonians

Write $V_X(\mathbf{r}) \equiv V(\mathbf{r}; \{l_{ij}^X\})$ for the potential with parameter set X . Three appear, and everything below turns on keeping them apart:

$$V_{\text{wt}}(\mathbf{r}) = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij}(\mathbf{r}) - l_{ij})^2 \quad (2)$$

$$V_{\text{mut}}(\mathbf{r}) = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij}(\mathbf{r}) - l_{ij} - \delta l_{ij})^2 \quad (3)$$

$$V_{\text{ref}}(\mathbf{r}) = \frac{1}{2} \sum_{i < j} k_{ij} (d_{ij}(\mathbf{r}) - d_{ij}(\mathbf{r}_{\text{mut}}^e))^2 \quad (4)$$

- (2) the wild type, springs l_{ij} .
- (3) the LFENM mutant: the mutation perturbs the natural lengths, $l_{ij} \rightarrow l_{ij} + \delta l_{ij}$, on the contacts of the mutated site. Nothing else changes. This Hamiltonian keeps the strain.
- (4) the refit: a *new* network built on the mutant's equilibrium conformation, with its springs reset to the distances found there. This is what “an ENM is a fit to a structure” produces, and it is the object SC-LFENM diagonalises.

For each, the associated quantities:

symbol	meaning
$\mathbf{F}_X(\mathbf{r}) = -\partial V_X / \partial \mathbf{r}$	the force
$\mathbf{r}_X^e$	the equilibrium conformation: where $\mathbf{F}_X(\mathbf{r}_X^e) = 0$
$V_X^{\min} \equiv V_X(\mathbf{r}_X^e)$	the minimum energy — a number, not a function
$\mathbf{e}_{ij}$	unit vector along the contact, $(\mathbf{r}_j - \mathbf{r}_i) / d_{ij}$
$\mathbf{K}_X$	the Hessian of V_X at $\mathbf{r}_X^e$; its null space is the six rigid-body motions
$\mathbf{C}_X = \mathbf{K}_X^+$	the covariance — Moore–Penrose pseudo-inverse of $\mathbf{K}_X$, inverting it on its range

The wild type is relaxed by construction

The ENM fit chooses $l_{ij} = d_{ij}(\mathbf{r}_{\text{wt}}^e)$ — every spring sits at its natural length in the native structure. Three consequences, used constantly below and all verified numerically:

$$\mathbf{F}_{\text{wt}}(\mathbf{r}_{\text{wt}}^e) = 0, \quad V_{\text{wt}}^{\min} = 0, \quad d_{ij}(\mathbf{r}_{\text{wt}}^e) - l_{ij} = 0 \quad \forall ij$$

The third is the one that matters most: the wild type carries no strain, and that is exactly what makes the cross term of §6 vanish.

The system. All numbers below are for acylphosphatase, 2acy chain A: 98 residues, 960 contacts at $d_{\text{max}} = 10.5$ Å, uniform $k_{ij} = 1$ (ANM), and mutation size $\sigma = 0.3$ unless stated otherwise. This is the structure every penm fixture is built from. Note that mutations perturb every contact of the mutated site — `mut_sd_min = 1`, not penm's default of 2, which excludes backbone neighbours.

2. The energy of a mutation

Step 1 — stress. Put $\mathbf{r} = \mathbf{r}_{\text{wt}}^e$ into the mutant Hamiltonian (3). Because the wild type is relaxed, $d_{ij}(\mathbf{r}_{\text{wt}}^e) = l_{ij}$ exactly, so the bracket collapses:

$$V_{\text{mut}}(\mathbf{r}_{\text{wt}}^e) = \frac{1}{2} \sum_{i < j} k_{ij} (l_{ij} - l_{ij} - \delta l_{ij})^2 = \frac{1}{2} \sum_{i < j} k_{ij} \delta l_{ij}^2 \equiv V_{\text{stress}}$$

strain erased

Figure 1: strain erased

Step 2 — the induced force. At that same conformation the mutant Hamiltonian is no longer at equilibrium; it exerts

$$\mathbf{f} \equiv \mathbf{F}_{\text{mut}}(\mathbf{r}_{\text{wt}}^e) = - \left. \frac{\partial V_{\text{mut}}}{\partial \mathbf{r}} \right|_{\mathbf{r}_{\text{wt}}^e}$$

Contact ij contributes a pair of equal and opposite forces of magnitude $k_{ij} \delta l_{ij}$ along $\mathbf{e}_{ij}$; the force on node i , written $\mathbf{f}_i$, is the sum over its contacts, and $\mathbf{f}$ stacks those N three-vectors.

Because these are central pair forces, $\sum_i \mathbf{f}_i = 0$ and the net torque about the centroid vanishes. That matters because $\mathbf{K}$'s null space is exactly the six rigid-body motions: a force with neither net force nor net torque is orthogonal to that null space, hence lies in the range of $\mathbf{K}$, and so $\mathbf{C}_{\text{wt}} \mathbf{f}$ is the true solution of $\mathbf{K} \delta \mathbf{r} = \mathbf{f}$ rather than the least-squares projection of an unsolvable system. Both checked numerically.

Step 3 — relaxation. To linear order the mutant's equilibrium conformation is

$$\mathbf{r}_{\text{mut}}^e \simeq \mathbf{r}_{\text{wt}}^e + \delta \mathbf{r}, \quad \delta \mathbf{r} = \mathbf{C}_{\text{wt}} \mathbf{f}$$

This is an *approximation*: the true $\mathbf{r}_{\text{mut}}^e$ satisfies $\mathbf{F}_{\text{mut}} = 0$, and the linear-response estimate leaves a residual force of order δl^2 (measured on 2acy: $|\mathbf{F}| = 2.1 \times 10^{-2}$, removed to 10^{-10} in 8 Newton iterations, moving the structure by 4×10^{-3} Å).

Step 4 — the energy change. The quantity of interest is the difference of *minimum* energies:

$$\Delta V = V_{\text{mut}}^{\min} - V_{\text{wt}}^{\min} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e)$$

and expanding about $\mathbf{r}_{\text{wt}}^e$ gives the two-term form,

$$\Delta V = \underbrace{\frac{1}{2} \sum k_{ij} \delta l_{ij}^2}_{V_{\text{stress}}} - \underbrace{\frac{1}{2} \delta \mathbf{r}^T \mathbf{K}_{\text{wt}} \delta \mathbf{r}}_{V_{\text{relax}}} + O(\delta l^3)$$

On 2acy, site 40: $V_{\text{stress}} = 0.7544$ and $V_{\text{relax}} = 0.1609$, so the decomposition gives $\Delta V = 0.5935$. Evaluating V_{mut} directly at the true minimum gives 0.5941 — a gap of 0.1 %, which is the $O(\delta l^3)$ truncation. $V_{\text{mut}}^{\min}$ is emphatically not zero: the mutant Hamiltonian keeps both the stress and the relaxation terms, exactly as in your `sc_lfenm.md`.

3. What the refit does, and why the energy must be saved

Now refit: build a *new* network on $\mathbf{r}_{\text{mut}}^e$, with springs $l_{ij} = d_{ij}^e(\mathbf{r}_{\text{mut}}^e)$. That is H_{ref} , and because its springs are relaxed by construction,

$$V_{\text{ref}}^{\min} = 0 \quad \text{while} \quad V_{\text{mut}}^{\min} = 0.594$$

Two different Hamiltonians share the structure $\mathbf{r}_{\text{mut}}^e$ and disagree about its energy. That gap is precisely the strain the refit threw away, and it is what has to be carried forward. (It is tempting to write $V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) = 0$ here. That is wrong: it attaches the refit Hamiltonian's energy to the mutant Hamiltonian's symbol, and collapses the whole distinction.)

error order

Figure 2: error order

three models

Figure 3: three models

How to read this. One mutation on 2acy. Bars 1 and 2 are V_{stress} and $-V_{\text{relax}}$; bar 3 their sum, the true ΔV . Bar 4 is $V_{\text{ref}}^{\text{min}}$ — what you get if you refit and then read the energy off the new network. Bar 5 restores the right answer by carrying the gap.

So each state stores a scalar offset `v_off`, and its energy is

$$V^{\text{min}}(\text{state}) = \underbrace{\frac{1}{2} \sum_{i < j} k_{ij} (d_{ij}^e - l_{ij})^2}_{\text{strain the network still carries}} + \text{v_off}$$

0 for the founder, correct for every descendant, and verified: refit energy plus offset reproduces $V_{\text{mut}}^{\text{min}}$ to machine precision. The offset is a constant — it never enters forces or K — so it is free dynamically. It is exactly the V_{ij}^{min} slot of your 2017 equation (1), which the current LFENM sets to zero and never uses.

One caveat on r^e in the code. A state stores one structure. For a refit state that structure is r^e exactly. For the non-refit models it is the linear-response estimate, carrying an $O(\delta l^2)$ residual force; `state_vmin(st, exact = TRUE)` relaxes to the true minimum first and is the default in `run_trajectory()`.

Every step above is checked numerically in `derivation.R`, which walks the derivation in order on 2acy site 40 — wild-type relaxedness, the stress energy, the force sum and torque, $K \delta r = f$, the Newton relaxation, the two-term identity, and finally $V_{\text{ref}}^{\text{min}} + \text{v_off} = V_{\text{mut}}^{\text{min}}$ — printing OK or DIFF for each of its 11 assertions. All pass.

4. The two-term formula is exact to $O(\delta l^3)$

$V_{\text{stress}} - V_{\text{relax}}$ is a quadratic approximation. Comparing it against direct evaluation of the perturbed potential at the relaxed mutant structure, over a 50-fold range of mutation sizes:

Left: the decomposition and the exact energy over a 50-fold range in σ ; both scale as σ^2 and the points sit on the line. Right: the discrepancy, on log axes, against a pure cubic (dashed). Fitted slope 2.95. The error is $O(\delta l^3)$, which is the linear-response truncation and nothing else; the bookkeeping itself is exact. At a realistic $\sigma = 0.3$ the error is 0.11 % at site 40 and 0.72 % at site 11 — it is site-dependent, so a single figure is not meaningful on its own.

Mutating site A then site B gives the same accumulated energy as B then A, to 6×10^{-4} — the same order as the truncation error. Note the scope: that test (`test_path_rebuild.R`) is on a synthetic 40-node helix, and its companion `test_path.R` covers the *non-rebuild* scheme, not SC-LFENM. Path-independence on 2acy under the rebuild is not established here.

5. Three models: **lfenm**, **keepnet**, **sclfenm**

“Rebuild or not” is not a single choice: there are three distinct models, and two of them are easy to confuse. They differ *only* in what happens to the Hessian K .

dV always positive

Figure 4: dV always positive

	what happens to the network	what happens to K	modes change?
lfenm	kept; l accumulates	frozen at the wild type	never
keepnet	kept; l accumulates	recomputed at the new structure	yes, through geometry
scfenm	rebuilt relaxed	rebuilt	yes, geometry + contact map

The rebuilt contact map really does differ, and by an amount that depends on the site: mutating site 40 takes 2acy from 960 contacts to 965, while the site-11 mutation of figure 6 adds 3. Wild-type and mutant are therefore networks over different edge sets, which is why sums over contacts need care when comparing them.

The middle one is the trap. “Don’t rebuild the contact list” does *not* mean “freeze K”: even with the same contacts, K drifts because the unit vectors e_{ij} and distances d_{ij} have moved. Textbook LFENM freezes K outright. The distinction is worth holding onto: the three give different answers for anything involving motion.

Figure 6 uses a mutation at site 11, so its energies are not the site-40 numbers of §§2–3. Note that the structure is identical in all three (second panel). All three relax with C_{wt} ; the difference appears only afterwards. And in the last two panels you can see the energy physically move out of the network and into the offset when you rebuild.

6. Re-referencing at every step makes the walk absorbing

This section diagnoses why the obvious way of running a trajectory fails. §8 gives the cure, so read the two together: the failure is real, but it is a consequence of *re-referencing*, not of SC-LFENM as such.

Under SC-LFENM, every mutation raises the energy. Without exception. In 400 random mutations from the wild type the smallest ΔV was +0.055; from a state already 30 substitutions along, +0.023 over 400 draws (fig3_dv_positive.R). Not one was negative.

Panel (a) shows the distribution of ΔV for 400 random mutations, starting from two states each 30 substitutions from the founder: a rebuilt SC-LFENM state (bottom, exactly relaxed, min $\Delta V = +0.023$) and a keepnet state carrying strain (top, min $\Delta V = -0.188$, 2 % of draws negative). The strained state has a tail crossing zero; the relaxed one does not come near it.

Why every mutation costs energy

A proof, not a demonstration. Let B be the $3N \times n_{\text{edge}}$ matrix whose column for contact ij holds $+e_{ij}$ in node i ’s three rows and $-e_{ij}$ in node j ’s — the incidence matrix weighted by contact directions — so that Step 2’s force is $f = -B(k \delta l)$. Define the weighted perturbation $u = \sqrt{k} \delta l$, one entry per contact. Then

$$V_{\text{stress}} = \frac{1}{2} \|u\|^2, \quad V_{\text{relax}} = \frac{1}{2} u^T M u, \quad M \equiv \sqrt{k} B^T C B \sqrt{k}$$

— the first from Step 1, the second by substituting $\delta r = C f$ into $\frac{1}{2} \delta r^T K \delta r$ and using $CKC = C$.

This is where relaxedness enters. The projector property depends on $K = B \text{diag}(k) B^T$, which holds *only* when every $g_{ij} = 0$ — that is, only for a relaxed network. Measured: on the relaxed wild type $\max |K - Bk B^T| = 0$ exactly; on a frustrated state it fails by 0.23 against a scale of 12.9. So “ $\Delta V \geq 0$ for any relaxed network” is not a rhetorical qualifier — the identity underpinning the proof is exactly what relaxedness buys.

trajectories

Figure 5: trajectories

M is an orthogonal projector. Verified on 2acy: $\max|M^2 - M| = 1.1 \times 10^{-14}$, every eigenvalue in $\{0, 1\}$, rank exactly $288 = 3N - 6$ — one unit eigenvalue per internal degree of freedom, since M 's range is the image of the non-rigid directions. A projector satisfies $u^T M u \leq \|u\|^2$ for every u , so

$$V_{\text{relax}} \leq V_{\text{stress}} \implies \Delta V \geq 0$$

with equality only when u lies entirely in the range of M — i.e. when the perturbation is fully absorbable by relaxation. Relaxation recovers at most the stress it was given, and the “at most” is a projection, not an inequality that needs checking.

This holds for any relaxed network — any structure, any cutoff — and is not special to the SC-LFENM refit. The refit merely guarantees relaxedness at every step. (`proof_projector.R` builds M explicitly, checks $M^2 = M$ and its spectrum, then confirms $V_{\text{relax}} \leq V_{\text{stress}}$ on 200 random single-site mutations.)

What strain buys

On a network that is *not* relaxed, expanding about its equilibrium conformation gives a third term:

$$\Delta V = \underbrace{\frac{1}{2} \sum k \delta l^2}_{>0, O(\sigma^2)} - \underbrace{\sum k (d_{ij}^e - l_{ij}) \delta l}_{\text{zero mean, } O(\sigma)} - \underbrace{\frac{1}{2} \delta r^T K \delta r}_{<0}$$

The cross term is proportional to the existing strain, and it is the only one that can be negative. The refit sets $d_{ij}^e = l_{ij}$ exactly, so under SC-LFENM it vanishes identically and $\Delta V > 0$ strictly.

Panel (b) of figure 3 shows the three contributions over 400 mutations of a strained keepnet state: the quadratic term always positive, the relaxation term always negative, and the cross term straddling zero with its mean on it.

The cross term has zero mean, so it adds variance, not drift. Worth stating plainly, because the natural reading — that strain gives keepnet access to strain-relieving mutations, lowering the energy on average — is wrong. Measured along a keepnet trajectory (`check_negative.R`, which walks 40 keepnet substitutions and at selected steps relaxes the state, measures its accumulated strain, then draws 250 fresh mutations from it):

substitutions	strain	mean ΔV	min ΔV	fraction < 0
5	2.7	+0.620	−0.079	0.008
20	15.6	+0.632	−0.464	0.040
40	73.8	+0.649	−1.597	0.104

The mean ΔV *rises* as strain accumulates. What widens is the distribution: from 0.8 % of mutations downhill at 5 substitutions to 10.4 % at 40. A strained network survives at an energy ceiling because its fatter left tail keeps producing the rare acceptable move — not because it drifts downhill.

One caveat on the argument: the cross term has zero mean over *unconditioned* δl draws, by symmetry of the Gaussian. Along a selected trajectory the draws are not unconditioned — acceptance prefers those making the cross term negative — so the unconditional statement does not by itself license a claim about a selected walk. The measured table above is the direct evidence, and it agrees.

What that does to a trajectory

Panel (a): energy accumulated from the founder, for five ways of running the same model. Grey is no selection at all. Purple and red are Metropolis on the step energy at two selection strengths — they *slow* the climb (slope 0.63 unselected, 0.56 at $\nu = 1$, 0.39 at $\nu = 4$) but cannot stop it, because there is no downhill move to find; even at $\nu = 4$, with 15 % of mutations accepted, the energy still rises steadily.

The two lines at the bottom are the interesting ones. Both impose a hard ceiling $V < 25$, which is what a stability criterion means when every step costs energy. Blue is SC-LFENM: it climbs to the ceiling and stalls at 44 substitutions (the $\times$) — no acceptable mutation exists any more, and the trajectory stops. Green is keepnet under the identical criterion, and it does not stall.

A comparison the figure alone does not license. Running keepnet out to 70 substitutions separates two regimes (`check_steady.R`, which repeats the ceiling run for 70 steps and reports mean energy, acceptance and drift in blocks of 20):

block	mean V	tries per acceptance	RMSD
steps 1–20	6.1	1.0	0.114
steps 21–40	17.4	1.0	0.210
steps 41–60	24.7	9.0	0.269
steps 61–70	24.9	13.5	0.310

For the first 40 substitutions acceptance is *free* — the walk is nowhere near the ceiling, and nothing is being tested. Only from step ~45 does the constraint bind. So a whole-run acceptance rate mixes unconstrained and constrained steps and is not comparable with a rate measured at the wall. The like-for-like figures are

keepnet at steady state: 0.123 vs SC-LFENM at its stall: 0.047

a factor of 2.6. The qualitative conclusion holds — keepnet reaches a steady state at the ceiling and keeps evolving (RMSD still climbing, $0.27 \rightarrow 0.31$, while V sits at 24.9), whereas SC-LFENM stops altogether — but the margin is a factor of two, not an order of magnitude, and it comes from the fatter left tail of the strained ΔV distribution, not from downhill drift.

Panel (b) of figure 4 plots RMSD from the founder for the same five runs. The blue SC-LFENM line simply ends at 44 substitutions; every other run keeps drifting, and the green one keeps drifting at constant energy — which is what a stationary evolutionary process should look like.

run	substitutions reached	outcome
SC-LFENM, no selection	50/50	energy runs away
SC-LFENM, Metropolis $\nu = 1$	50/50	energy runs away, slower
SC-LFENM, Metropolis $\nu = 4$	50/50	energy runs away, slower still
SC-LFENM, ceiling $V < 25$	44/50	stalls
keepnet, ceiling $V < 25$	70/70	stationary at the ceiling; still drifting structurally

This is a consequence of the design, not a bug. If the network is refit to relaxed at every step, the model has no memory of strain, and therefore no way to release any.

scanning

Figure 6: scanning

response matrix

Figure 7: response matrix

7. Where SC-LFENM does work: scanning

For single-mutation scanning the problem above never arises, because every mutant is measured against a *fixed* founder and nothing accumulates. Here the rebuild is pure gain: it gives you the changed spectrum that LFENM cannot produce at all.

Run on 2acy chain A (98 residues, 960 contacts at $d_{\max} = 10.5 \text{ \AA}$).

(a) Energy cost of mutating each site, averaged over 15 mutations, with spread. (b) That cost against contact number: $r = 0.97$ — buried sites cost more, the standard stability result, on a real structure. (c) How far the mutated site itself moves, against contact number: $r = -0.70$ — buried sites move less, as they must, being held by more springs. (d) The change in conformational entropy: identically zero under LFENM, which cannot see it at all; range $[-0.35, +0.29]$ under SC-LFENM.

Response and influence are different things

The natural object is the response matrix

$$R_{ij} = \langle |\delta r_i|^2 \rangle \quad \text{averaged over mutations at site } j$$

— the response *at* site i to mutating site j .

Panel (a) is the matrix itself on a log scale: a bright diagonal (a site responds most to its own mutation), bands along the contacts, and off-diagonal ridges where sequence-distant segments are in spatial contact. Panel (b) shows the two marginals against burial.

It has two marginals, and they answer different questions:

marginal	name	meaning	vs contact number
$\sum_j R_{ij}$	response of site i	how much i moves when the protein is mutated anywhere	$r = -0.80$
$\sum_i R_{ij}$	influence of site j	how much the protein moves when j is mutated	$r = +0.20$

Both computed analytically (no sampling) in `response_influence.R`; the two marginals sum to the same total, 14.777, checked three ways.

They point opposite ways and both are right. A buried site is held by more springs, so it responds less. But mutating a buried site forces all of its many contacts, so it influences more. The diagonal R_{jj} — how far the mutated site itself moves — follows the *response* correlation, not the influence one: figure 5(c) plots it against contact number and gives $r = -0.70$.

The two marginals are easy to confuse, and confusing them inverts the sign. An independent check settles which is which, using no mutation at all: equilibrium mean-square fluctuation against contact number gives $r = -0.80$, matching the *response* marginal exactly. Buried sites fluctuate less; buried sites respond less; these are the same statement.

R is not symmetric, and should not be. `prs_theory.md` notes that the single-edge response matrix is symmetric — stretching one spring is one spring’s worth of perturbation however you label its ends. But a *site* mutation forces all cn_j contacts of j , so column sums scale with contact number while row sums do not: measured asymmetry $\max |R - R^T| / \max R = 0.49$, identical in the analytical and simulated matrices, so it is structural rather than sampling noise.

8. Trajectories made stationary: the catalogue model

§6 showed that re-referencing the network at every step makes every mutation uphill, so the walk is absorbing. The fix is not to re-reference. It is yours, and it works.

The idea

Keep one fixed reference — the founder. For each site j , catalogue a set of alternative states $m = 0, 1, \dots, M$, where $m = 0$ means “not mutated” (this is already penm’s convention: `mutation = 0` returns the wild type unchanged). For each, record the change *relative to the founder*:

$$X(0 \rightarrow j, m) \quad \text{for } X \in \{\Delta V, \delta r, \Delta(TS), \dots\}$$

A move between two states of the same site is then a difference of catalogued quantities:

$$X(j, m_0 \rightarrow m_1) = X(0 \rightarrow j, m_1) - X(0 \rightarrow j, m_0)$$

Be precise about what this is and is not. As algebra on one site’s stored values it is a triviality — $X_{m_1} - X_{m_0} = (X_{m_1} - X_0) - (X_{m_0} - X_0)$. It is exact for the physical energy only when the rest of the protein is in the state the catalogue was built against, i.e. the founder. Used mid-trajectory, when other sites have already mutated, it inherits a site-additivity approximation (quantified below). So the identity defines a *surrogate* energy

$$V_{\text{cat}}(s) = \sum_j \Delta V(0 \rightarrow j, s_j)$$

and the trajectory is exact in V_{cat} , approximate in V .

That distinction granted, three consequences follow, and they are what make the scheme work:

1. The bookkeeping is exactly antisymmetric: $X(m_1 \rightarrow m_0) = -X(m_0 \rightarrow m_1)$ in V_{cat} , so the walk is reversible by construction. This is a property of the ledger, not a claim about the physical energy — but it is the property a Markov chain needs.
2. ΔV_{cat} takes both signs — negative whenever m_1 is a cheaper state of that site than m_0 . (Quoting “50 % of *ordered* pairs are downhill” would be a tautology, forced by antisymmetry. The informative statement: over 450 unordered pairs at 30 sites the median $|\Delta V|$ is 0.16 and the maximum 1.01, so a downhill direction exists for essentially every pair of states.)
3. The state space is finite and the energy has a floor. $(M + 1)^N$ sequences, and every catalogue entry is ≥ 0 (measured range $[0, 1.647]$), so $V_{\text{cat}} \geq 0$ with the founder at the bottom.

Consequence 3 is the operative one, and it is worth being explicit that §6’s theorem is not violated here but respected. §6 proves $\Delta V \geq 0$ for any mutation measured from a *relaxed* network’s own minimum. The founder is relaxed, so every catalogue entry is indeed non-negative. What changed is that the reference is now fixed: the floor stays at the founder instead of moving up to meet the walker at every step. Downhill moves exist because a site can revert toward a cheaper state — partially back toward the founder — not because any single mutation from a relaxed reference has become negative.

catalogue

Figure 8: catalogue

Two changes were made at once

The catalogue is built with LFENM, not SC-LFENM (penm_catalog.R uses mut_model = "lfenm"). So §8 differs from §6 in two ways — a fixed reference *and* a different model — and the cure should not be credited entirely to the first. The fixed reference is what gives the finite state space and the floor; LFENM is what makes the catalogue’s structural entries background-independent (§8, “The one assumption”). Both are needed.

Why LFENM makes this exact

penm’s calculate_force() computes $f_{ij} = -k_{ij} \delta l_{ij}$ directly from the perturbation, and calculate_dxyz() returns $\delta r = C_{wt} f$. That is the linearisation the name “linearly forced” refers to: expand to first order in δl , and the mutation becomes an external force on the unchanged wild-type network.

Because δr is *linear* in f , a trajectory $f_1, f_2, \dots$ gives

$$\delta r_{\text{total}} = C \left(\sum_i f_i \right) = \sum_i C f_i = \sum_i \delta r_i$$

Structure is exactly additive, and therefore exactly reversible. Verified on 2acy to machine precision:

identity	measured
$\delta r(\sum f) = \sum \delta r$	4.5×10^{-15}
forward then exact reverse returns the founder	$ \delta r = 7.5 \times 10^{-17}$
$\Delta V(1 \rightarrow 2) + \Delta V(2 \rightarrow 1) = 0$	0 exactly
mutation = 0 returns the wild type	identical

It works

(a) The distribution of ΔV between states of a site, over 4400 ordered pairs at 40 sites. It is symmetric about zero by antisymmetry, so the 50 % figure is forced; what the panel shows is the *spread* — median $|\Delta V| = 0.16$, tails to ± 1.6 . Compare §6, where every mutation from the current state was uphill and no downhill move existed at all.

(b) Trajectories under a Metropolis rule on the catalogue difference, at four selection strengths. Every one reaches a stationary state: the energy climbs from the founder, then plateaus and fluctuates about a stable mean. No stalling, no runaway. The plateau level is set by selection.

(c) The equilibrium energy against the analytical prediction $\langle V_{\text{cat}} \rangle = \sum_j \sum_m p_{jm} \Delta V_{jm}$ with $p_{jm} \propto e^{-\nu \Delta V_{jm}}$. Agreement to 0.6–4 % over a 32-fold range in ν .

What this does and does not show. V_{cat} is separable by construction, and the proposal is symmetric, so a correct Metropolis sampler *must* reach the product measure $\prod_j p_{jm}$ — panel (c) is a unit test of the sampler, not evidence about the physics. It cannot fail if the code is right.

The question it does not answer is whether the *exact* energy of the sampled sequences behaves the same way. Running the chain on V_{cat} and then evaluating the true V of the sampled sequences:

ν	predicted	$\langle V_{\text{cat}} \rangle$	$\langle V_{\text{exact}} \rangle$	difference
1	47.6	48.5	49.5	−2.2 %
2	39.0	38.1	37.9	+0.5 %

ν	predicted	$\langle V_{\text{cat}} \rangle$	$\langle V_{\text{exact}} \rangle$	difference
5	16.2	14.1	14.2	-1.1 %

The surrogate tracks the exact energy to about 2 % in the mean (`test_exact_boltzmann.R`), so the chain samples something close to $e^{-\nu V}$ — but “close to”, measured, not “provably”.

Convergence is from both directions: starting at the founder ($V = 0$) the walk rises to $\langle V \rangle = 36.4$; starting from a random sequence ($V = 61.8$) it falls to 34.9. Same level from above and below.

The one assumption, and how it degrades

The difference identity is exact within V_{cat} . What is *assumed* is that a site’s states have background-independent effects: the catalogue is built against the founder but used when other sites have already mutated.

Measured on one protocol for both models — take eight fixed perturbations of site 40, apply them to backgrounds evolved 5 to 40 substitutions away, and compare against their effect on the founder (`test_background_final.R`):

substitutions	$\text{cor}(\Delta V)$	ΔV error	$ \delta r $ error
5	0.968	9.2 %	0.66 %
10	0.876	19.8 %	6.3 %
20	0.868	21.7 %	8.7 %
40	0.607	48.0 %	17.3 %

The error does grow. An earlier measurement suggested otherwise; it ran to only 20 substitutions on a different protocol. The structural response is very good early (0.66 % at 5 substitutions, as the linearity argument predicts, since C is frozen) but degrades as the accumulated δl moves the geometry away from the founder. So a catalogue built once against the founder is reliable for tens of substitutions, not hundreds, and for long trajectories it should be rebuilt periodically against the current state.

The energy error is *entirely* the cross term of §6: correlation between the background-dependence of ΔV and the computed cross term is 0.9999, residual $\sim 10^{-3}$ (`test_background3.R`). Since the cross term is computable from the current state’s strain, it can be corrected for — the cheapest way to extend the catalogue’s useful range.

SC-LFENM cannot be catalogued at all. The rebuild changes the contact set (960 $\rightarrow$ 959 edges after five substitutions), so a δl vector built on the founder no longer matches the mutant’s graph; the operation is not merely inaccurate but undefined. This is a stronger reason to use LFENM for the catalogue than the error magnitudes.

A second assumption: additivity across sites

The catalogue energy of a sequence sums single-site contributions, $V(s) = \sum_j \Delta V(0 \rightarrow j, s_j)$. This is a *different* approximation from the difference identity — that one is exact; this one assumes mutations at different sites do not interact. Measured against the exact energy of the combined perturbation:

sites mutated	catalogue	exact	relative error
1	0.752	0.752	0.0 %
5	3.060	3.022	1.2 %
20	11.61	11.85	-2.2 %
98 (all)	60.66	61.38	-1.1 %

Within about 2 % even when every site is mutated, and not growing systematically with the number of sites (`test_site_additivity.R`).

Per move the error is much larger, and can flip a sign. Aggregate agreement hides it. On a background with 30 sites mutated, comparing the catalogue's ΔV for a single move against the exact value (`test_catalog_honest.R`, 12 moves):

site	move	catalogue	exact	error
33	10 $\rightarrow$ 8	-0.401	-0.426	0.025
14	8 $\rightarrow$ 9	-0.040	-0.155	0.116
31	5 $\rightarrow$ 4	+0.232	-0.048	0.280
23	1 $\rightarrow$ 10	+0.136	+0.015	0.122

One move in twelve had the wrong sign — the catalogue called it uphill when it was in fact downhill. So the chain is making some accept/reject decisions on the wrong side of zero. The aggregate energy is still tracked to 2 %, because these errors are unbiased and average out, but a study that cared about *individual* substitutions rather than ensemble properties would need the exact ΔV at each step.

9. With and without frustration

A network is frustrated when $l_{ij} \neq d_{ij}^e$: its springs are stressed at the equilibrium conformation. The word covers two separate questions, and conflating them causes most of the confusion around the `frustrated` flag.

(A) Does the model let a network be frustrated? — a question about $\{l_{ij}\}$. (B) Does the Hessian keep the transverse term? — a question about K .

The wild type is never frustrated

`set_enm()` fits $l_{ij} = d_{ij}$, so on any freshly built network $\max|l - d| = 0$ exactly and the flag is a strict no-op: $\|K_{\text{frust}} - K_{\text{relaxed}}\| = 0$. This is why `frustrated = FALSE` costs nothing for a wild type, and why the blocked `stopifnot(!frustrated)` has never been felt.

Mutation creates frustration

The moment a mutation is applied, $l_{ij} \rightarrow l_{ij} + \delta l_{ij}$ while the structure relaxes only partially, so the two part company:

substitutions	$\max l - d $	strain energy
1	0.58	0.65
3	0.64	1.59
6	1.02	2.82

So every LFENM mutant is frustrated, exactly as your 2017 document argues (“as we introduce mutations frustrations are introduced”). The question is whether that frustration is allowed to reach the dynamics.

A trap worth recording

The frustrated Hessian must be evaluated at a stationary point. The LFENM structure is the linear-response estimate, not the true minimum — on a 10-mutation state the residual force is $|F| = 0.61$. A Hessian taken there is not singular on rotations, and produces three spurious eigenvalues of order 4×10^{-4} that masquerade as broken rotational invariance. They are not physical: the potential is built from the scalars d_{ij} and l_{ij} , so it is rotationally invariant by construction and its Hessian *must* have six zero modes. Relaxing to the true minimum (5 Newton iterations, $|F| \rightarrow 8 \times 10^{-14}$) restores exactly six.

frustration

Figure 9: frustration

Why three and not six. Translations are exact zero modes at *any* conformation, because V depends only on the differences $\mathbf{r}_j - \mathbf{r}_i$, so a uniform shift is annihilated identically. Rotations are zero modes only *via* $\mathbf{F} = 0$: rotating a configuration by ω leaves V unchanged, but the second-order term in the expansion carries $\sum_i \mathbf{F}_i \cdot (\omega \times (\omega \times \mathbf{r}_i))$, which survives when $\mathbf{F} \neq 0$. Hence exactly three spurious modes off a stationary point, and none at the minimum. If those modes are left in, TS is wrong by 8.6 — the entropy sum contains $\log(1/\lambda)$, which diverges as $\lambda \rightarrow 0$.

How much does the transverse term actually change?

With all spectra taken at the true minimum:

(a) The Hessians do differ, and the gap grows with accumulated strain — from 0.6 % at one substitution to 2.7 % at forty. (b) But the soft modes barely rotate: RMSIP over the ten softest modes stays above 0.998 throughout. (c) The entropy difference is small and takes both signs, $|\Delta(TS)| < 0.11$.

The lowest non-rigid eigenvalue stays positive at every strain level reached (≈ 0.20 throughout), so these frustrated networks are genuine minima, not saddles. That bounds a worry raised in §6's earlier drafts: severe frustration *could* drive an eigenvalue negative, but at the strains an LFENM trajectory actually reaches, it does not.

Reading. Frustration is real, it is created by every mutation, and it changes the Hessian by a measurable but small amount. For structural quantities it is irrelevant; for entropy differences of order 0.05 it is at the edge of significance. The `frustrated = TRUE` branch is not needed for what `penm` currently computes — but the reason is measured, not assumed.

10. Which K for the response? — an open question from 2020, answered

NOTES.Rmd (2020-05-14) asks: “*Rigorously, should I use K_{wt} or K_{mut} to calculate δr ?*” It has stood unanswered. It is cheap to test: compare `penm`’s one-step $\delta r = C_{wt}f$ against relaxing to the true minimum of the mutant’s own potential.

	$ \delta r $ error	direction (cos)
8 random sites, $\sigma = 0.3$	1.3 – 10.4 %	0.9946 – 0.9999

The direction is right to better than one part in a hundred; only the magnitude is off, and by a few percent. The error scales linearly with mutation size — 1.2 % at $\sigma = 0.1$, 9.4 % at $\sigma = 0.8$ — exactly as an $O(\delta l^2)$ correction should.

So K_{wt} is the right choice for practical purposes, and the question can be closed: the self-consistent response points the same way and differs only in length, by an amount controlled by σ .

11. What this says about `penm`’s own **sclfenm**

The package blocks `frustrated = TRUE` and its `sclfenm` tests skip, both with notes saying the handling was never checked. That is the reason for this exploration, so it is worth reporting what the exploration finds about the package code itself (`test_penm_sclfenm.R`).

Three things that are fine

check	result
<code>mutate_graph()</code> 's <code>lij</code> copy, which carries “ <i>true only if edges are ordered</i> ”	The two <code>%in%</code> selections align pairwise (958 rows each, labels matching). Correct here; the warning concerns a case that does not arise, because <code>%in%</code> on a sorted edge list preserves order.
the mutant's <code>dij</code> against its own <code>xyz</code>	exact
the mutant's <code>eij</code> against its own <code>xyz</code>	exact — <code>mutate_enm()</code> calls <code>set_enm_eij()</code>

One real problem: the SC-LFENM mutant is not at a minimum of its own network

Measured as the residual force of the mutant evaluated on its own graph:

	$ F $
LFENM mutant	2.2×10^{-2}
SC-LFENM mutant	7.1×10^{-1}

a factor of 33. The cause is that `get_mutant_site_sc_lfenm()` takes one linear step with C_{wt} , then rebuilds the graph around the resulting structure — but never re-relaxes the structure against the *new* graph. So r_{mut} is a minimum of neither network.

And the effect is entirely driven by lost contacts. Over 20 random sites:

contact set	mean $ F $ ratio (sc_lfenm / lfenm)
unchanged	1.00
changed	4.08, with excursions to $21\times$

The asymmetry is the mechanism. A gained contact arrives with $l_{ij} = d_{ij}$ by construction, exerts no force, and costs nothing — site 42 gains two edges and its ratio stays exactly 1.00. A lost contact was carrying strain (measured: 0.22 and 0.27 Å for the two edges site 8 loses), and deleting it removes a force the structure had equilibrated against, leaving the remainder out of balance.

This matters because §9 shows that spectra taken off a stationary point are wrong in a specific way — three spurious near-zero modes, and TS off by 8.6. Any entropy or mode quantity computed from penm's `sc_lfenm` mutant inherits that error. The fix is one line of iteration: after `mutate_enm()`, relax the structure against the rebuilt graph until $F = 0$ (`relax_to_min()` here does it in about five Newton steps). I have not made that change to the package.

12. Recommendations

Scanning: SC-LFENM with the offset. Well behaved, energetics exact to third order, and the only one of the three models that gives mutational effects on dynamics (§7).

Trajectories: the catalogue (§8). Build the catalogue once against the founder, then evolve by picking a site and a new state and accepting on the catalogue difference. It is exactly reversible, it reaches a stationary state, and it samples $e^{-\nu V}$. Two practical notes:

- Use LFENM for the catalogue, not SC-LFENM. SC-LFENM's rebuild changes the contact set, so a founder-built δl does not even match the mutant graph — the catalogue is undefined, not just inaccurate (§8).

- Rebuild the catalogue periodically. Background-independence degrades with distance from the founder: the structural error runs 0.7 % at 5 substitutions to 17 % at 40 (§8). For trajectories of more than a few tens of substitutions, re-catalogue against the current state, or correct for the cross term, which accounts for essentially all of the energy error.
- Rebuild only to observe. If you want mutant modes or entropies along a trajectory, rebuild a relaxed network at the current structure at that moment and diagonalise it — do not evolve with the rebuild in the loop. This is the “fast” option your 2020 NOTES.Rmd describes, and the catalogue makes it clean: the evolutionary bookkeeping and the spectral measurement are separate operations on the same state.

The offset scheme is worth keeping regardless of which of these you use. It is what makes energy well defined across a rebuild.

What is still open. The catalogue’s states are drawn from a Gaussian δl with no amino-acid identity. If you want states that mean something — a 20-letter alphabet, or l_{ij} depending on the residue pair — that is the natural next step, and the catalogue structure accommodates it directly: replace “state m of site j ” by “residue type a at site j ” and nothing else changes. Your 2017 appendix lists this as strategy 2.

13. What I checked, and how you can re-run it

The five checks

Rscript validate.R mutates site 11 of 2acy under each model and prints PASS/FAIL for five assertions.

check	what it asserts	live?	result
V1	the total energy after a rebuild, offset included, equals the energy without the rebuild	live	PASS (agree to 10^{-10})
V2	ΔV from the model equals $V_{\text{stress}} - V_{\text{relax}}$ to better than 1 %	live	PASS (0.72 %)
V3	LFENM and SC-LFENM produce the same mutant structure	no	PASS (RMSD exactly 0)
V4	$\Delta(TS)$ is zero under LFENM, non-zero under SC-LFENM	half	PASS
V5	the founder has zero energy	no	PASS

Three of the five cannot fail as written, and that is a defect. V3 compares two structures that `mutate()` assigns from the same variable, so they are equal by construction whatever the code does. V5 asserts $0 = 0$. V4’s LFENM half compares a frozen spectrum with itself; only its SC-LFENM half can go red. Those three document intent rather than test it.

To confirm the live checks *can* fail, `test_can_fail.R` breaks the code deliberately: zeroing the stored offset makes the reported energy 0 instead of 0.626, and flipping the sign of the Hessian’s transverse term makes the numerical-derivative check miss by 4.96 instead of 0.00104.

The Hessian convention

The transverse term is $g_{ij} = l_{ij}/d_{ij} - 1$ paired with the bracket $(ee^T - I)$. This looks like the opposite of the textbook form, and is not: the two sign flips cancel,

$$\left(\frac{l}{d} - 1\right) (ee^T - I) = \left(1 - \frac{l}{d}\right) (I - ee^T)$$

so it agrees with $\partial^2 V / \partial r_i \partial r_j = -k[ee^T + (1-l/d)(I-ee^T)]$. Checked against a numerical Hessian on strained networks, where the genuinely wrong sign is off by $O(1)$.

It defaults to off. A relaxed network has $g = 0$ either way, so this can only matter for keepnet — which is where §6's argument lives, so it needed checking rather than assuming. Measured on a 30-step keepnet state (strain energy 15.5, $\max |d - l| = 1.27$): the two Hessians differ by 2.3 %, the ten softest eigenvalues by 1.1 %, and ΔV over 150 mutations is unchanged (means 0.5688 vs 0.5686, correlation 1.00000). Harmless here, and measured rather than asserted (`check_frustrated.R`).

Files

Organised by role — see `README.md` for the full map.

folder	contents
R/	the library: <code>enm_core.R</code> , <code>sclfenm.R</code> , <code>applications.R</code> (used by §§1–7); <code>penm_catalog.R</code> , <code>frustration.R</code> (used by §§8–11); <code>fig_setup.R</code>
figures/ checks/	one script per figure, beside its <code>.png</code> scripts that assert and can fail: <code>validate.R</code> , <code>derivation.R</code> , <code>test_identities.R</code> , <code>test_can_fail.R</code> , <code>proof_projector.R</code> , <code>test_stationary.R</code> , <code>test_two_impls.R</code>
analyses/	scripts that compute the numbers quoted here: <code>run_*.R</code> , <code>test_background_final.R</code> , <code>test_site_additivity.R</code> , <code>test_catalog_honest.R</code> , <code>test_exact_boltzmann.R</code> , <code>test_penm_sclfenm*.R</code> , and others
attic/ data/	superseded diagnostics, kept for provenance regenerable <code>.rds</code> artifacts (gitignored)

All scripts use the here package, so they run from any working directory.

The Hessian convention

The transverse term is $g_{ij} = l_{ij}/d_{ij} - 1$ paired with the bracket $(ee^T - I)$. This looks like the opposite of the textbook form, and is not: the two sign flips cancel,

$$\left(\frac{l}{d} - 1\right) (ee^T - I) = \left(1 - \frac{l}{d}\right) (I - ee^T)$$

so it agrees with $\partial^2 V / \partial r_i \partial r_j = -k[ee^T + (1-l/d)(I-ee^T)]$. Checked against a numerical Hessian on strained networks, where the genuinely wrong sign is off by $O(1)$.

It defaults to off. A relaxed network has $g = 0$ either way, so this can only matter for keepnet — which is where §6's argument lives, so it needed checking rather than assuming. Measured on a 30-step keepnet state (strain energy 15.5, $\max |d - l| = 1.27$): the two Hessians differ by 2.3 %, the ten softest eigenvalues by 1.1 %, and ΔV over 150 mutations is unchanged (means 0.5688 vs 0.5686, correlation 1.00000). Harmless here, and measured rather than asserted (`check_frustrated.R`).

Files

file	what it is
enm_core.R	the ENM itself: potential, force, Hessian, normal modes, relaxed fit
sclfenm.R	the model — <code>state</code> , <code>mutate()</code> , <code>state_vmin()</code> , <code>state_req()</code> , structure and motion comparisons
applications.R	<code>scan_sites()</code> and <code>run_trajectory()</code>
validate.R	the five checks in the table above
test_can_fail.R	deliberate sabotage; confirms the checks detect it
fig1...fig9_*.R	the nine figures, each self-contained
figs/	the rendered figures
test_*.R, diag_*.R	the individual numerical experiments behind §4 and §6
response_influence.R	the response matrix, computed analytically, and its marginals
penm_catalog.R	the catalogue model, built on penm's own functions
test_identities.R	the four exact identities the catalogue rests on
test_stationary.R	convergence from both directions, detailed balance, reversibility
test_site_additivity.R	how well the catalogue energy sums across sites
test_catalog_honest.R	catalogue vs exact ΔV for single moves on an evolved background
test_exact_boltzmann.R	does the chain sample $e^{-\nu V}$ for the <i>exact</i> V ?
test_background_final.R	background-independence for both models, one protocol
test_background2.R, test_background3.R	background-independence, and that its error is the cross term
frustration.R, run_frustration*.R	with vs without frustration; includes <code>relax_to_min()</code>
run_which_K.R	K_{wt} vs K_{mut} for the response
test_penm_sclfenm*.R	what this exploration says about penm's own <code>sclfenm</code> (§11)
test_two_impls.R	that the reimplementations and penm agree to machine precision
check_req.R, check_req2.R	checks that the stored structure really is r^e , and how much V is off when it is not

14. Limitations

- One protein (2acy chain A, 98 residues), one model (ANM, uniform k), one cut-off (10.5 Å), one mutation size ($\sigma = 0.3$). The identities of §8 are algebraic and hold generally; every number would change on another system.
- The catalogue's states are Gaussian draws, not amino acids. There is no alphabet and no sequence-dependent l_{ij} (§11).
- Sites are treated as independent in the catalogue energy: within 2 % in aggregate even with all 98 sites mutated, but per-move errors can flip a sign (§8). This is a real approximation, unlike the difference identity.
- §4's path-independence number (6×10^{-4}) is from a synthetic 40-node helix, not 2acy — the one result here not measured on the real protein.
- The catalogue is built with LFENM. Using it with SC-LFENM observations assumes the two agree on structure, which they do exactly (§5), but the entropies then come from a network the catalogue did not price.

- Trajectory lengths here are 10^3 – 10^4 proposed mutations at a single temperature; no attempt at a proper phylogeny or at branch-length calibration.